

# IN-FLIGHT AND GROUND INSTRUMENTATION FOR DAUPHIN'S FENESTRON NOISE INVESTIGATION

**Laurent Binet**, ONERA, Center of Salon de Provence, France

**Fabrice Cuzieux**, ONERA, Center of Salon de Provence, France

**Jean Claude Camus**, CEV, Istres, France

**Laurent Perthuis**, CEV, Istres, France

## ABSTRACT

The Fenestron had been developed, conceived and produced as an alternate solution to the conventional tail rotor in order to improve security and performance for rotorcraft, and was used for the first time at the end of the 1960s. The advantages of the Fenestron are numerous in terms of safety aspects, aerodynamic efficiency and noise reduction. Therefore, many rotorcrafts in the world use the Fenestron as tail rotor, for light, intermediate and medium weight helicopters. Nevertheless, even if this concept has demonstrated high performances for noise reduction, acoustic emissions could still be reduced in some peculiar flight configurations. The French Directorate for Civil Aeronautics Programs (DGAC) has funded a Fenestron Noise Study Program in order to understand and analyze the aero-acoustics phenomena involved in these configurations. This program is based on previous research dedicated to the development and the validation of ONERA's aerodynamic/acoustic computational tools for the analysis and prediction of noise generated by a Fenestron.

The French Flight Test Center (CEV) at Istres and Eurocopter were involved in this research program which was conducted by the ONERA's Department of Computational Fluid Dynamics and Aeroacoustics.

Flight tests with the research helicopter Dauphin 6075 of the French Flight Test Center at Istres have been preferred to wind tunnel tests which are unable to reproduce realistic in-flight conditions.

The French Flight Test Center, Eurocopter and several departments of ONERA have been in charge of designing and building the aft equipment which was integrated and tested in flight on the helicopter.

During a first campaign, onboard aerodynamic and acoustic measurements were performed to identify the flight conditions where the Fenestron is noisiest. A second test campaign began in March 2010 in order to measure helicopter acoustic footprint on the ground and to identify correlations between onboard and ground measurements in the noisiest flight conditions previously identified.

The resulting experimental database will help to understand how the sound is generated by a helicopter Fenestron as well as to validate the aeroacoustic computation codes from ONERA which could be used to improve flight procedures and/or the shape of the Fenestron for noise reduction.

## ACRONYMS

|       |                                                         |
|-------|---------------------------------------------------------|
| DGAC  | French General Directorate of Civil Aviation            |
| CEV   | French flight test Center                               |
| ONERA | Office National d'Etudes et de Recherches Aérospatiales |
| HOST  | Helicopter Overall Simulation Tool                      |

## 1. INTRODUCTION

The French Directorate for Civil Aeronautics Programs (DGAC) has funded a Fenestron Noise Study Program in order to understand and analyze the aero-acoustics phenomena responsible for the noisiest flight conditions regarding the Fenestron.

The Flight Test Center (CEV) at Istres and Eurocopter were involved in this research program conducted by the Department of Computational Fluid Dynamics and Aeroacoustics of ONERA.

A first flight test campaign started in 2008 to study the sources of the noise nuisance generated by the Fenestron rotor on the Dauphin 365N helicopter and the flight conditions for which the Fenestron is the noisiest.

Eurocopter has already looked at Fenestron noise reduction by introducing into the rotor design a change in the blade distribution, a reduction in the circumferential speeds, and thin-blade stators. This led to the successive development of three generations of the Fenestron tail rotor.

Even if the Dauphin 365N-6075 is equipped with a first generation of the Fenestron, the resulting experimental database will help to validate the aeroacoustic computation codes from ONERA which could be used to improve flight procedures and/or the shape of future generations of Fenestron.

Previously planned to begin in September 2009, a second flight test campaign was dedicated to measurements of the helicopter acoustic ground footprint. Due to onboard instrumentation modifications and repairing delays after the first campaign, it had to be postponed to March 2010. As a matter of fact, results of this campaign, available when writing this paper, were not sufficiently consistent for being discussed here.

After a brief presentation of the first flight test campaign onboard instrumentation, an overview of the second test campaign instrumentation and its dedicated flight test procedures is presented.

Then, this paper will focus on the analysis of relevant results issued from the first test campaign. As detailed later the innovative onboard instrumentation installed on the 365N helicopter allowed us to study the influence of temperatures in the Fenestron duct highlighting some interactions with exhaust gases, depending on flight conditions.

Aerodynamic measurements are also discussed and compared with some HOST flight mechanic code results.

Finally, in-flight blade strains are analyzed in the entire flight domain explored as well as pressure measurements underlying a correlation between unsteady pressures and Fenestron noise generation.

## 2. BRIEF DESCRIPTION OF THE ONBOARD INSTRUMENTATION

This part of the paper deals with the on-board instrumentation implemented on the Dauphin 6075 of the CEV.

In order to perform onboard aerodynamic and acoustic measurements, a very innovative instrumentation and measurement chain architecture was achieved by Eurocopter, the CEV and the Department of System Controls and Flight Dynamics of ONERA.

The Fenestron blades, duct, fuselage sides and tail of the Dauphin 6075 helicopter (Figure 1) have been equipped with this advanced instrumentation.

In a previous paper, a complete description of the instrumentation installed on the helicopter has already been presented [1].



Figure 1: Dauphin 6075

### 2.1. Fixed instrumentation

For the fixed instrumentation chain, a data acquisition subsystem (DAS), on the shelf, from ACRA Control has been used.

Sensors used in the fixed part of the helicopter were:

- 4 microphones Brüel & Kjaer 1/4" mounted on the empennage, 2 on each helicopter side to perform acoustic measurements
- 13 pressure holes on both sides of the helicopter tail boom to measure the steady pressure around the rear part of the fuselage
- 4 generating lines of the duct were fitted each with 20 steady pressure sensors, all flush mounted
- 8 unsteady pressure transducers (Kulite sensors) mounted into the generating lines and positioned upstream and downstream of the blade plan (Only 4 Kulite sensors were finally operational because the integration of these sensors was very delicate and some of the wires have broken when they were stuck on the fuselage).
- 3 equally spaced radial rakes in the Fenestron duct. Each of them carrying 5 pitot tubes providing the local kinetic pressure and one 5-holes probe providing in addition the measurement of the local incidence and sideslip angles
- 1 synchronization pulse
- 5 mini electronic pressure scanners of 32 channels each receiving the signals from the static pressure measurements (rakes, rigid and pressure holes) and referenced to the helicopter static pressure.

Figure 2 shows the different sensor implemented around and in the Fenestron duct.

## 2.2. Rotating instrumentation

Getting measurements from sensors installed on a rotor turning at a nominal rotation speed close to 4,700 RPM is a real challenge.

In order to understand the physical mechanisms responsible for Fenestron noise generation, a dedicated instrumentation was implemented on the rotor blades to measure unsteady pressures. For this purpose, 10 upper/lower surface thin layer unsteady pressure transducers Kulites LL-072 were distributed

on 4 of the Fenestron blades, also providing temperatures.

To monitor in real time in flight the structural efforts on the blades, 3 strain gauges were also mounted on one blade at different radius as shown in Figure 3.

Hence, sensors installed on five blades of the tail rotor delivered the following measurements:

- Pressures
- Temperatures
- Strains

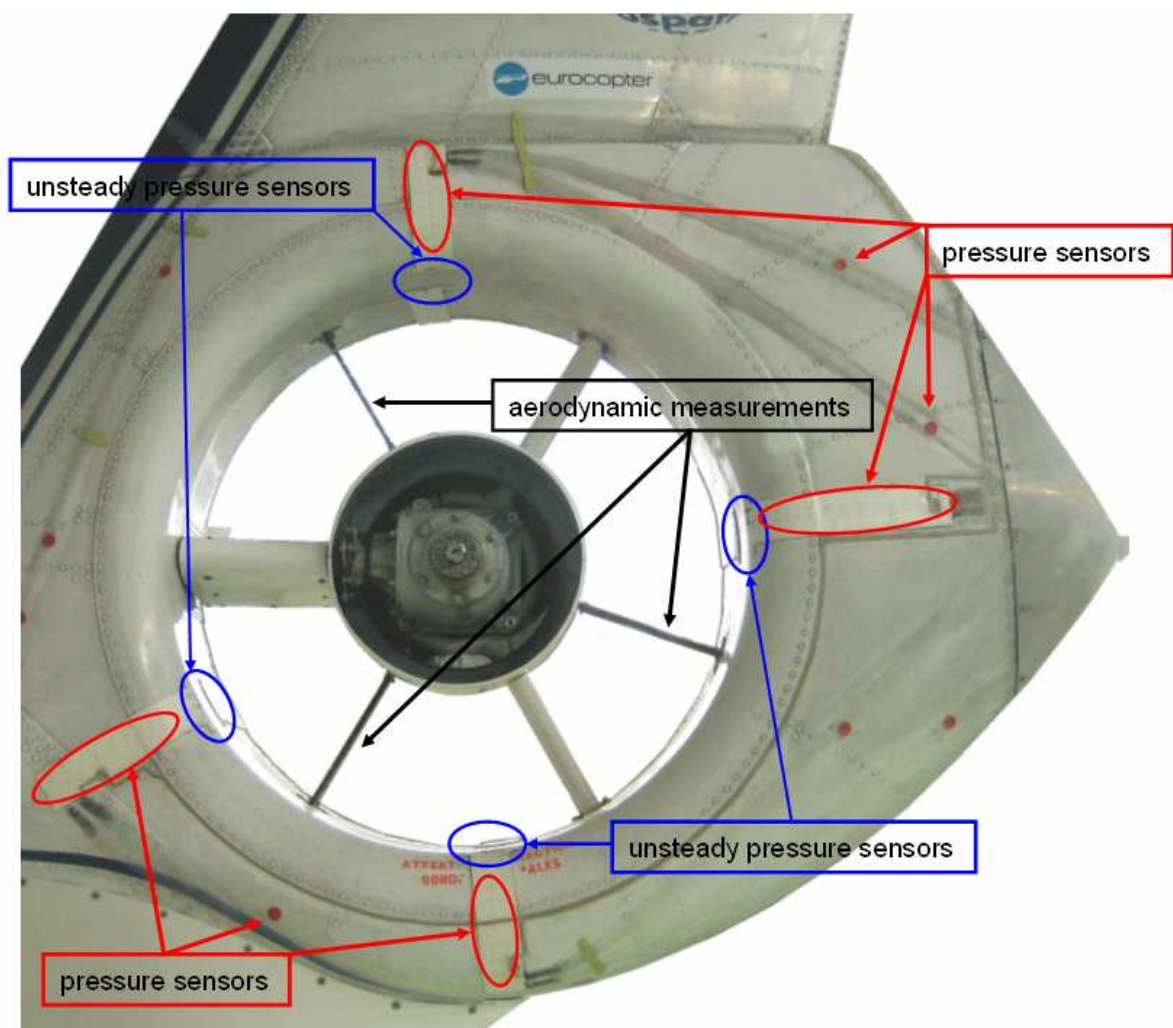
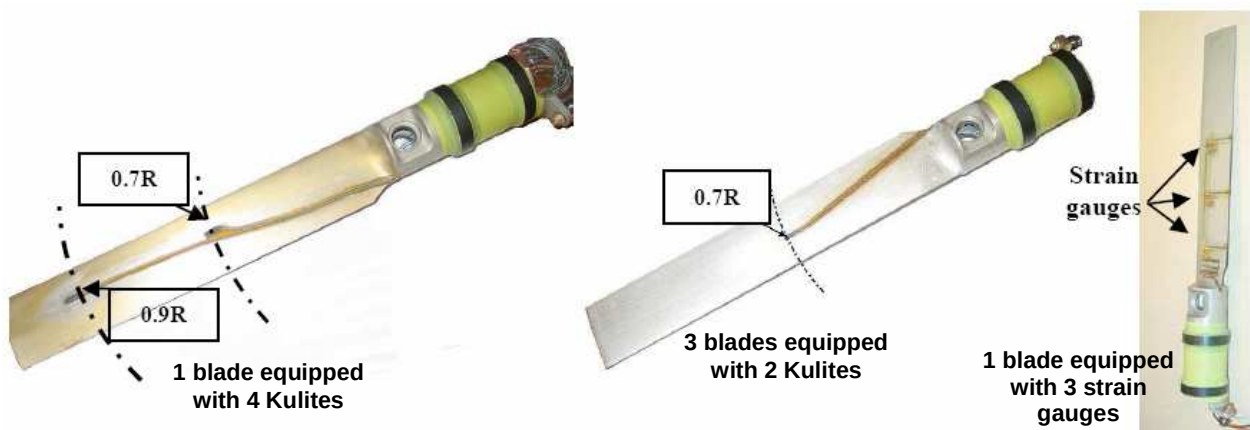


Figure 2: Instrumentation in fixed part



**Figure 3: Instrumented blades**

Moreover, a fast-rate telemetry was designed and built by DeKerac Company to transmit all the measurements performed in the rotating rotor to a high-capacity recorder installed in the helicopter cabin.

This fast rate telemetry has three major functions:

- Sensors conditioning (power supply, amplification, filtering);
- Acquisition of numerical signals (sampling, digitalizing and multiplexing) ;
- Radio transmission of the signals to the cabin receiver

Figure 4 shows two pictures of this fast rate telemetry. On the upper part, the telemetry placed on the centre, its power supply and support can be seen. Below, are indicated the three different electronic parts of the in-rotor telemetry system.

It is important to mention that, turning at a nominal RPM of around 4700, the electronic components were submitted to very high centrifugal forces. In the figure 4, the centrifugal forces as induced by the rotor rotation speed at two different radius are also indicated. Indeed, the centrifugal acceleration increases with respect to radial location at a rate of 247g/cm.

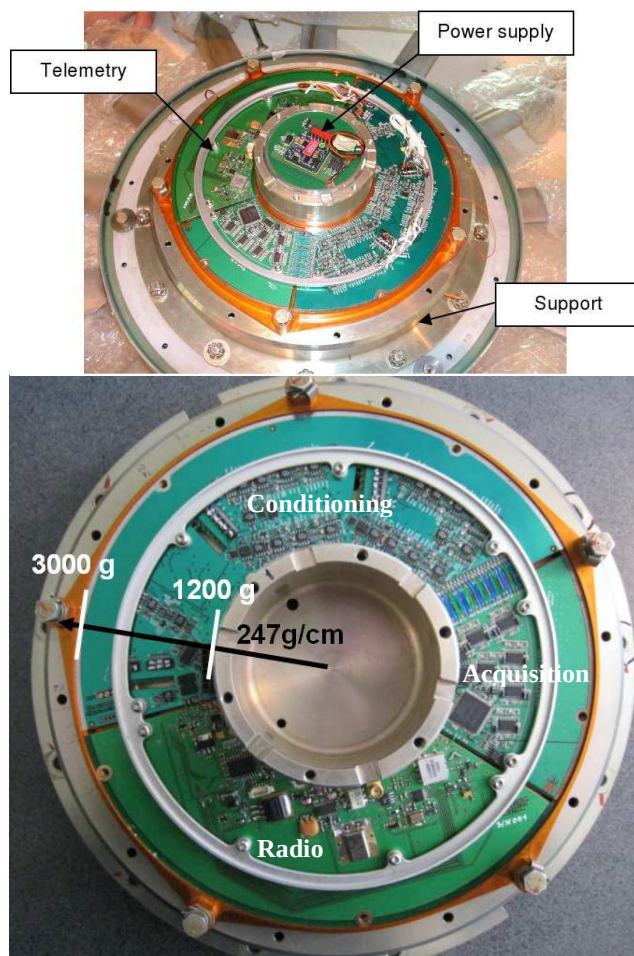
Moreover, a LED and a photoreceptor were fixed on the rotor to provide synchronizing pulses between fixed and rotating measurement chains.

A common time base for measurements performed in the rotating and non-rotating frames was used.

### 3. GROUND INSTRUMENTATION

In order to measure helicopter acoustic footprint on the ground and to identify correlations between onboard and ground measurements, a specific instrumentation will be installed on the ground for the second flight test campaign.

The definition of the measurement device is based on the DLR/ ONERA study "RONAP" [3] for which flight tests were conducted in order to perform acoustic measurements on the ground.



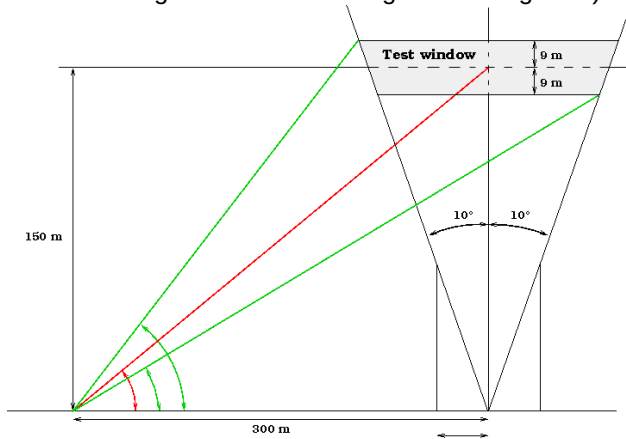
**Figure 4: On-board telemetry**



This device will be composed of a 500 meters length line on which are uniformly distributed 11 microphones Brüel & Kjaer of 1/2" diameter.

This line will be placed perpendicular to the trajectory of the helicopter and centered on its trajectory.

By considering the flight specifications used for noise overflight certification measurements which were provided by EUROCOPTER, the angle of reception of the most off-centered microphones will vary between 23,5° and 30,3° (values corresponding to the angle between the green lines and the ground on figure 5).



**Figure 5: Angles of reception of the off-centered microphones**

In order to know the relative position and orientation of the helicopter and the microphone, the position of the microphones will be precisely geo-positioned.

### 3.1. Microphones

The ONERA 1/2" microphone supports (Fig. 6) were manufactured for previous studies. The interest of such a mounting is detailed in paper [3]. In this case, microphones are fixed 6 mm above the plate for which reflected and incident noise are identical.

The plates, of about 45 cm diameter, ensure good reflection of the highest frequencies. The lower frequency noise reflection being ensured by the surrounding area which has to be as flat as possible, and of high impedance.



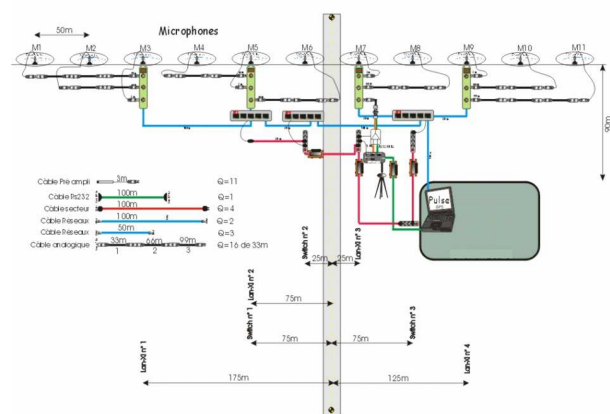
**Figure 6: ONERA microphone supports**

### 3.2. Ground measurements systems

A new acquisition chain Brüel & Kjaer will be used. It is composed of 3 modules of 4 channels each. This set enables to implement 11 microphones and to keep a channel for the record of the GPS time. All the modules are connected to a laptop on which the data will be recorded.

The ground measurements systems will be placed near a small building enabling us to store the different elements and providing the power supply.

The general diagram of the device is presented here after (Fig 7):



**Figure 7: ONERA noise measurements systems**

The microphones will be calibrated each measurement day morning using a 134 dB - 250 Hz pistonphone. The background noise will be also measured.

In order to avoid as far as possible the problem of humidity and, moreover, the possible destruction of cables caused by animals, this outdoor device will be dismantled at the end of each measurement day.

## 4. FLIGHT TESTS

### 4.1. First flight test campaign

A flight test program was defined by ONERA and Eurocopter (Figure 8) in order to identify the flight conditions when the Fenestron is more particularly noisiest.

This unique instrumentation induced specific constraints which were taken into account in the flight tests.

- Strain gauges (static and dynamic strains) had to be monitored in real time during the flights.
- Specific procedures: The lowest airspeed was fixed to 50 kts and conventional take-off and

landing procedures have been abandoned in favor of a rolling take-off between 0 and 50 kts.

- Specific instructions were established in order to allow the crew to minimize the blade loads (safe flight domain) and to land in the best conditions of safety.
- The in-rotor batteries needed for the telemetry system had 1h20min of autonomy which had an impact on the duration of the flights.
- Specific meteorological constraints had also to be taken into account:
  - o Low humidity and no rain.
  - o The outside temperature had to be lower than 30°
  - o Atmospheric perturbations, wind and gusts had to be low as much as possible
- Due to the time needed to analyze the loads obtained in flight on the blades and the time needed to obtain the next flight authorization, only one flight per week was possible.

The test program was compounded of different flight phases: take-off, approach phase, level flight, side slip flight.

Selected flight cases were repeated in order to validate the good agreement of the results between different flights. Thus, the flight case 75 kts and -800 ft/min was repeated at each flight as a reference case.

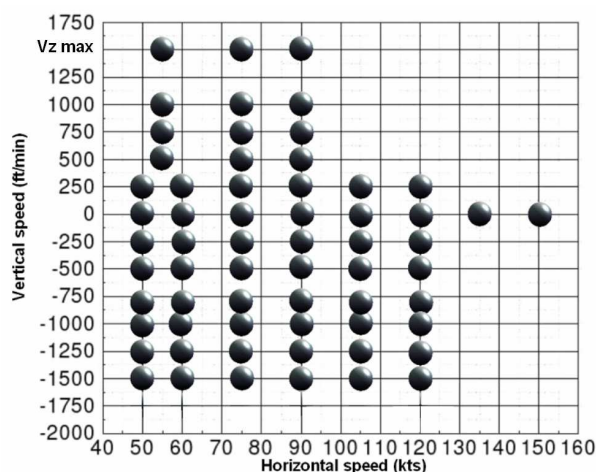


Figure 8: Flight test program of the first campaign

#### 4.2. Second test campaign

As this will be detailed in a following chapter, some sensors failed during the first flight test campaign requiring repairs for the second campaign.

Moreover, some modifications were done on the fixed instrumentation. Indeed, the three measurement rakes placed in the Fenestron duct have been removed. This modification was planned from the beginning of the study in order to verify that, positioned in the Fenestron duct in an intrusive way, they were not responsible of parasite noise.

Furthermore, LED and photoreceptor wires of the synchronization pulses so suffered of the high centrifugal forces that their fixations were modified. Finally, two specific devices were elaborated to calibrate the tail rotor torque and the blades pitch angle, enabling the measurement of these parameters during next flights.

Thus, previously planned to begin in September 2009, the second test campaign finally started at the end of March 2010.

Dedicated to the measurement of the helicopter acoustic footprint on the ground, the second test campaign will be performed at the French Air Force base of Istres. Of course, noise measurements outside an Air Force base are possible but, fenced and guarded 24 h/day, this has the advantage of reducing the risk for ground-based acoustic equipment to be stolen or damaged between test days. But on the other hand, Istres is a large multi-role tasked French Air Force base with several operational units and some test facilities which will add specific constraints, especially with regard to the background noise. The heavy traffic on Istres will be a major constrain during noise measurements and regulation of the air traffic will be necessary to limit the background noise as far as possible.

A Flight Director based on the Dauphin onboard instrumentation will be designed in order to display the trajectory to the pilot. The device will be based on the computation of heading (track) and slope angle deviations from reference points located on the path.

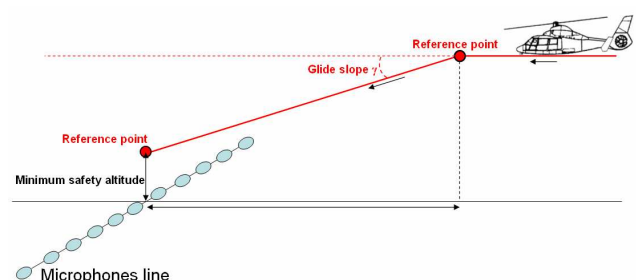


Figure 9: Approach scenario

Figure 9 presents an approach scenario. In this case, the two reference points will be defined for each of the 2 segments composing the trajectory. The pilot task

will be to intercept and to maintain a glide path up to the second reference point above the microphones line. For that goal, the position of the center of the microphones line will be precisely determined.

The glide path indication will be given to the pilot via a specific instrument onboard the Dauphin-6075 as presented in figure 10. The crossed needles on this display are used to give the helicopter slope angle and heading. The objective for the pilot will be to keep the needles intersection at the screen centre in order to be on the targeted glide path and flight direction. The needles are scaled to give an indication of the precision on each parameter.

Reprogramming this flight director in flight is impossible, so each measurement day will be dedicated to a specific procedure: approach, level flight or take-off.

Due to time constraints and this second flight test campaign being more dedicated to the ground noise measurements, only the first flight will be performed with the instrumented tail rotor. A classical tail rotor will then be used to finish the second test campaign, allowing several flights in a week.



**Figure 10: Dauphin on-board instrumentation**

Based on the results of the first campaign, a flight test program has been defined and is presented here after (Table 1). It is composed of 24 flight cases corresponding to level flights, approaches, take-off and approaches with sideslip.

Flight cases with sideslip at 90kt and -800 ft/min were not flew in the initial campaign but first acoustics analyses revealed that this operating point corresponds to a transition region between low and high OASPL (Over All Sound Pressure Levels). Loading the Fenestron by flying with a sideslip angle has demonstrated noise reduction ability [5] which should be confirmed in these tests.

| Level Flight : |       |       |        |        |        |        |
|----------------|-------|-------|--------|--------|--------|--------|
| 60 kt          | 75 kt | 90 kt | 105 kt | 120 kt | 135 kt | 150 kt |
| X              | X     | X     | X      | X      | X      | X      |

| Approach:   | 50 kt | 75 kt | 90 kt | 105 kt |
|-------------|-------|-------|-------|--------|
| -500 ft/mn  |       |       |       | X      |
| -800 ft/mn  |       | X     | X     |        |
| -1000 ft/mn | X     | X     | X     | X      |
| -1250 ft/mn |       | X     | X     |        |
| -1500 ft/mn | X     | X     |       | X      |

| Take-off : | 55 kt | 75 kt | 90 kt |
|------------|-------|-------|-------|
| Vz max     | X     | X     | X     |

| Approach with sideslip: | +5° | -5° |
|-------------------------|-----|-----|
| 90 kt / -800 ft/mn      | X   | X   |

**Table 1: Flight test program of the second campaign**

## 5. RESULTS

### 5.1. First flight test campaign

The first campaign has been carried out from November 2008 to January 2009 and the entire flight test program was successfully achieved in eight flights and 11 flight hours.

Some sensors failed during the flights but the overall instrumentation gave very satisfactory results.

The unique and innovative features implemented on the Dauphin and the successful achievement of the first flight campaign provided a fairly complete aerodynamic and acoustic database relative to the flow and noise mechanisms in which the Dauphin Fenestron is involved.

### 5.2. Onboard Instrumentation

The fixed instrumentation worked perfectly during all flights. Only a few pressure sensors or taps turned out to be unserviceable and one pitot tube mounted on an aerodynamic measurement comb came to be obstructed.

The outcome has to be tempered regarding the instrumentation installed on the rotating parts. The fast rate telemetry perfectly worked but due to the very high centrifugal constraints, some problems occurred with different connecting wires.

Indeed, in the first flights dedicated to the determination of the flight test envelope, the synchronizing signal mounted on the rotor was lost. It was rapidly discovered that the wire between the telemetry and the photoreceptor was broken.

The loss of this signal in rotating instrumentation chain unfortunately prevented the possibility of synchronizing the two measurement chains. As discussed after, another way to count the tail rotor RPM was observed but it was not sufficiently precise to be used to synchronize fixed and rotating measurement chains.

Some measurements on the blades were also progressively lost during the flights, connecting wires being teared out by the centrifugal forces. Due to time constraints repairing was impossible during the test campaign, and finally only four Kulite sensors situated on two blades were still correctly working at the end of the flight tests.

### 5.3. Experimental data

ONERA departments of Aerodynamics and Aeroacoustics are analyzing aerodynamic and acoustic results and this database will be very helpful to validate aero-acoustic computation codes.

However, some measurements have been used by the Department of System Controls and Flight Dynamics to study how the Fenestron works in terms of dynamic loads and to improve the flight mechanics models.

On 132 flight cases performed during the campaign, 223 stabilized phases from 2 to 4 seconds have been extracted, constituting a rather large database for the following analysis.

#### 5.3.1. Temperatures in the Fenestron duct

The pressure transducers mounted on the blades and in the Fenestron duct are temperature compensated to correct the output for errors induced by the change in bridge resistance due to temperature. Therefore, the local temperatures on the blades, at the bottom and at the rear part of the Fenestron duct were used in the post-treatment of the pressure sensors and the analysis of these parameters has been initiated. First results revealed a variation of the temperature inside the Fenestron vein and on the blades depending on the flight case. Indeed, even if the engine outlets are at about 4 meters from the Fenestron, it appears that the exhausted gas can be ingested by the tail rotor under certain conditions.

Some computations have been undertaken with the shared EUROCOPTER flight mechanics code HOST to evaluate if the ingestion of warm gas in the

Fenestron could change its performances. Table 2 summarizes these calculations.

| Flight cases                    | Additional Fenestron power needed | Additional total required power needed | Additional pilot command (pedals) needed |
|---------------------------------|-----------------------------------|----------------------------------------|------------------------------------------|
| Level flight 75 kts             | +1655 W                           | + 0.37%                                | 54.08%→55.97%<br>+ 1.89%                 |
| Hover OGE                       | +2840 W                           | + 0.37 %                               | 77.8%→78.9%<br>+1.1 %                    |
| Descent 75kts<br>-<br>800ft/min | +77.8 W                           | +0.02%                                 | 46.77%→47.34%<br>+0.57 %                 |

**Table 2: Impact of the temperatures on the Fenestron performances**

Three flight cases have been analyzed: level flight at 75 kts, hover OGE and a descent flight at 75 kts and - 800 ft/min. In each case, the temperature in the Fenestron duct has been modified to fit with the experimental data.

As it can be seen, the impact on the performances is very low. More power is needed by the Fenestron and the total required power increases, but in very small proportions. The pilot has to compensate with a higher pedal command but, here again, the differences are small.

Experiments show that depending of the flight case, the local temperature in the Fenestron duct can vary but computations have shown that the impact on the performances remains relatively small.

Nevertheless, if we assume that the exhaust gases are responsible for the variation of the temperature in the Fenestron duct, it seems that these variations should be dependant with the wake angle of the exhaust gases. In order to prove this assumption, a model based on a mixed use between HOST simulation results and the experimental data available is under development.

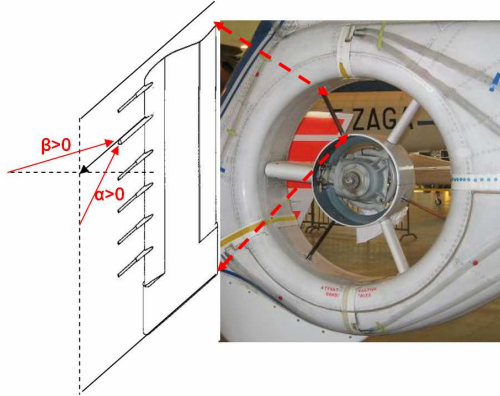
Despite many assumptions and estimated parameters (exhaust gases velocity, mean rotor induced velocity, etc), first results are promising and such a modeling of the interaction between exhaust gases and the Fenestron would improve flight mechanic codes.

#### 5.3.2. Aerodynamic measurements in the Fenestron duct

As shown in the next figure (Figure 11), the 3 measurement rakes were placed very close to the tail



rotor plan. As represented on the left of the figure, the 5-holes probes mounted on the rakes provided the measurement of the local incidence ( $\alpha$ ) and side slip ( $\beta$ ) angles.



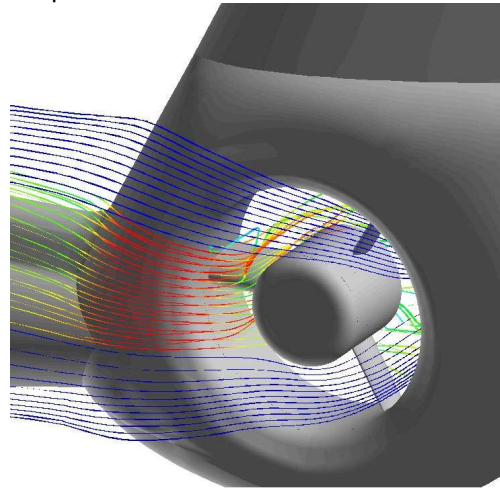
**Figure 11: Incidence and sideslip angles provided by the 5 holes probes**

In a previous ONERA paper [4], CFD computations revealed the complexity of the flow in the Fenestron duct. Taken from this paper, the following picture (Figure 12) is a close view at the 3D streamlines in the Fenestron vein showing a strong influence of the tail rotor drive support, and an inhomogeneous flow field in the vein.

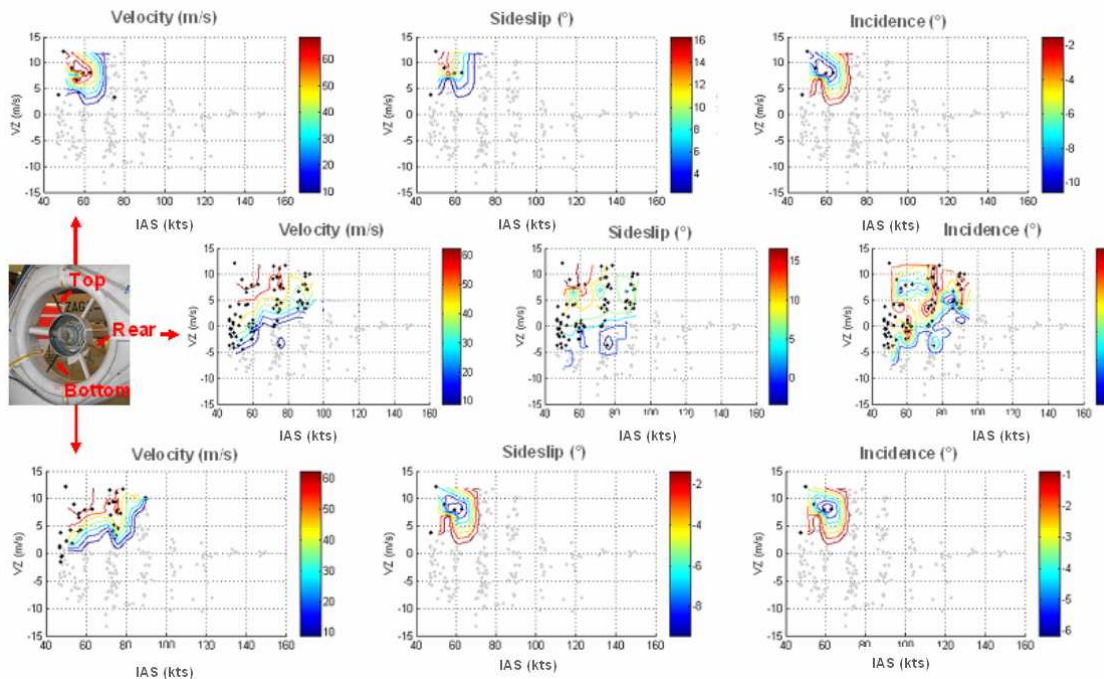
Moreover, pitot tubes and 5 holes probes had a measurement range of  $\pm 25^\circ$  around their centre line. For incidence and sideslip higher than  $\pm 25^\circ$ , meaning

that the flow direction is deflected from the probe axis, measures were unavailable.

In Figure 13 are represented the velocities, the local incidence and sideslip obtained on the three measurement rakes in the entire first campaign. Flight phases during which such measurements were available are not numerous. Available data were mainly observed at low Indicated Airspeed and in climb, when the loads on the rotor are high, confirming a very disturbed airflow in the duct as foreseen by CFD computations.



**Figure 12: CFD results - Streamlines in the Fenestron in level flight**



**Figure 13: Airflow measurement provided by the 5 holes probes**

At high airspeeds, very few measurements were available, suggesting that the probes were badly positioned and/or that the speed of the flow was too small to be measured. This low level of speed in the flow can be explained by the increase of efficiency of the fin at high airspeeds. Then, the tail rotor loading and the induced velocities are highly reduced in these conditions.

In conclusion, the region where probe signals were significant corresponds to climb at moderate speeds. Depending on the location of the probes, some measures were also available in level flight or moderate descent but always at low speeds. No velocities were measured for Indicated Airspeed higher than 95 kts.

In spite of its location just after the rotor hub, the rear rake provided measurements on a larger domain than the other rakes. The rake placed at the bottom of the vein was efficient only for positive vertical speeds. The one located at the top only for very steep climbs.

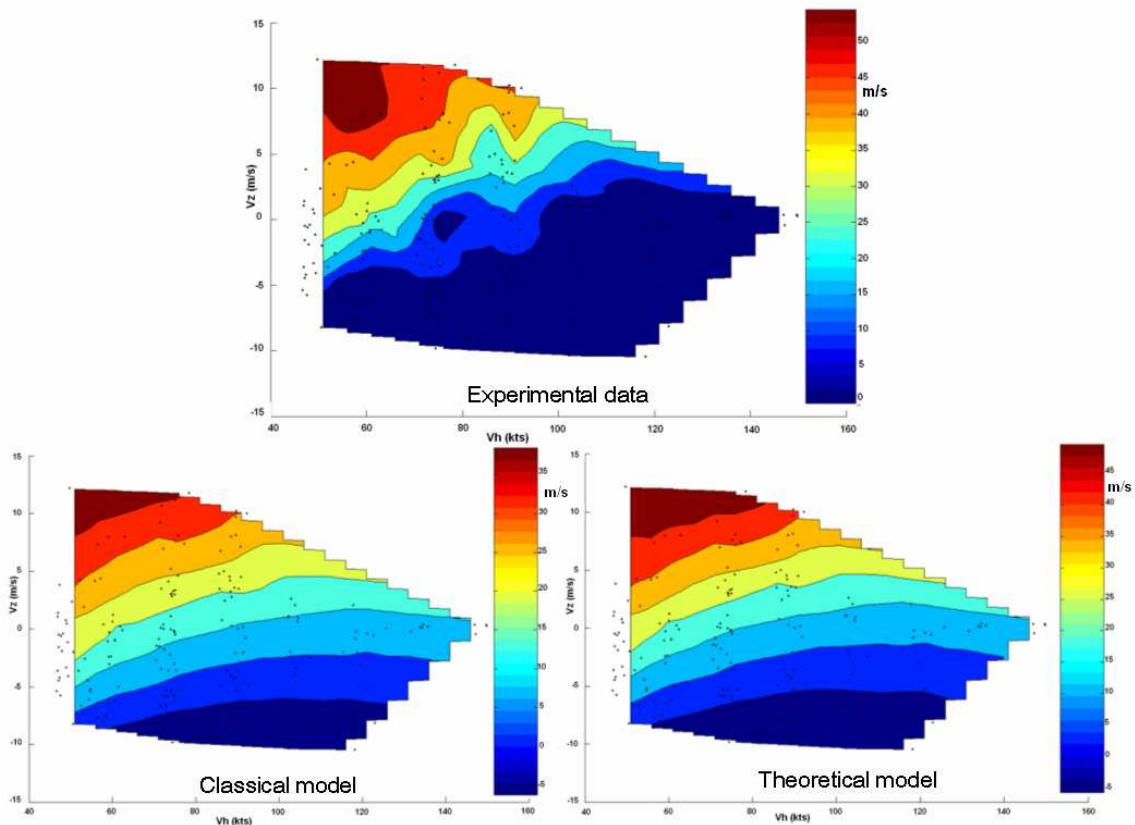
Maximal values are almost the same for the three probes and equal to 77.13 m/s on the bottom rake, 74.39 m/s on the rear rake and 78.8 m/s for the top

rake. All these values were recorded during climb phases.

Analysis and comparisons between airflow measurements and a flight mechanics code have been performed and are represented here after. Experimental velocities have been compared to two Fenestron models implemented in HOST (figure 14).

The classical tail rotor model is based on a standard tail rotor adapted to a ducted propeller by controlling the flow contraction, the drag in advancing flight and no blade flapping. The theoretical model is an adaptation where some parameters have been adjusted to match wind tunnel data on a 1/2.8 scale of a Fenestron model.

On the upper part of the figure is represented the average values of the velocities measured during all stabilized phases by the 5 holes probes. Due to their threshold and measurement ranges, a large domain of this chart indicates null values. This graph uses an interpolation algorithm providing values between 0 and 10 m/s but the minimal experimental velocity is equal to 10.25 m/s.



**Figure 14:** Comparison between experimental velocities of the three 5 holes probes in a  $V_h/V_z$  diagram and computed mean induced velocities obtained with the different Fenestron models of HOST code

The distribution of the velocities is relatively the same for both HOST models. As expected, velocities are the highest in climb at moderate forward speed. These velocities gradually decrease with the vertical and forward speeds. Nevertheless, both models underestimate this decrease. Moreover, the highest values predicted by both models are quite lower than measured in flight. The theoretical model is better and calculates a maximal velocity of 57.7 m/s instead of 62.9 m/s in flight.

Contrary to the probes which were oriented upstream, HOST models are able to provide negative induced velocities in steep descent. The fin providing an antitorque force during forward flight regimes, the Fenestron thrust is then highly reduced. In steep descent, the Fenestron is then off-loaded, working in a "reverse" mode to counteract the aerodynamic loads applied on the fin and the horizontal stabilizer side fins. This reverse mode has been clearly observed by analyzing the unsteady pressures on the blades and has been detailed a previous ONERA paper [2].

In order to evaluate the impact of a modification of the induced velocity on the performances, the classical HOST model has been modified and tested on a selected flight case. The forward speed is 49.77 kts, rate of climb is 12.17 m/s. Table 3 shows the impact of the increase of the induced velocity to match with experimental data.

|                | Fenestron power needed (KW) | Total required power needed (KW) | Pilot command (pedals) needed (%) | Induced velocity (m/s) |
|----------------|-----------------------------|----------------------------------|-----------------------------------|------------------------|
| Initial model  | 148.84                      | 980.39                           | 83.83                             | 45.274                 |
| Modified model | 205.99 (+ 38.39%)           | 1039.1 (+ 5.98%)                 | 93.9 (+ 10%)                      | 62.9                   |

**Table 3: Impact of the modification of the Fenestron induced velocity on the performances in climb**

The initial Fenestron model gave 45.274 m/s of induced velocity. Once modified to give 62.9m/s, it can be seen an increase of the Fenestron power needed, the total required power needed and the pilot command needed. To perform this climb phase, the total required power increases of 5.98%. The command needed increases of 10%, getting closer to the end stop.

At the beginning of the test campaign, hover flights were performed for the on-board instrumentation validation.

These flights have not been included in the test matrix as some parameters were unknown already available but, same computations have been made to evaluate the impact of an increase of the Fenestron induced velocity in hover. Results are presented here after (Table 4).

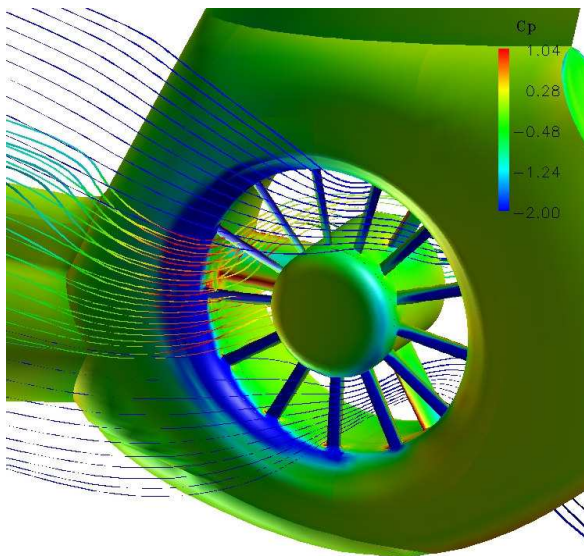
|                | Fenestron power needed (KW) | Total required power needed (KW) | Pilot command (pedals) Needed (%) | Induced velocity (m/s) |
|----------------|-----------------------------|----------------------------------|-----------------------------------|------------------------|
| Initial model  | 64.705                      | 654.44                           | 80.449                            | 59.57                  |
| Modified model | 76.498 (+ 18.22%)           | 667.23 (+ 1.95%)                 | 86.474 (+ 6%)                     | 70.4                   |

**Table 4: Impact of the modification of the Fenestron induced velocity on the performances in hover**

As for climb phases, an increase of the induced velocity in hover leads to an increase of the total required power but in smaller proportions. The same remarks can be made on the command needed.

Flight phases when the three 5 holes probes were simultaneously measuring flow velocity as well as the local incidence and sideslip are rare and were mainly observed in climb phases.

Figure 15 represents, on the lower part, the velocity vectors plotted in the Fenestron frame for all these phases and on the upper part, a CFD computation showing the 3D streamlines in the Fenestron vein. Contrary to the rear rake (green vectors) which mainly captured the induced velocity of the rotor, including a swirl effect, both front rakes (blue and red vectors) are more influenced by the fairing effect. In the lower part of the figure are also plotted the velocity vectors obtained by CFD computations in a vertical cut corresponding to the plan of the probes.



**Figure 15: CFD computation compared to velocity vectors on the three 5 holes probes**

The blades generate a helical flow field downstream which is relatively well measured by the rear rake (black and green vectors). Nevertheless, combined to the incoming flow, this helical field is modified and disappears in the bottom part of the Fenestron vein (red and black vectors). Under these conditions, the real inflow in the Fenestron vein is far from an ideal axial column, and skewed inflows are presented to the blades. Finally, as already illustrated in Figure 12, streamlines underline also a large influence of the tail rotor drive support.

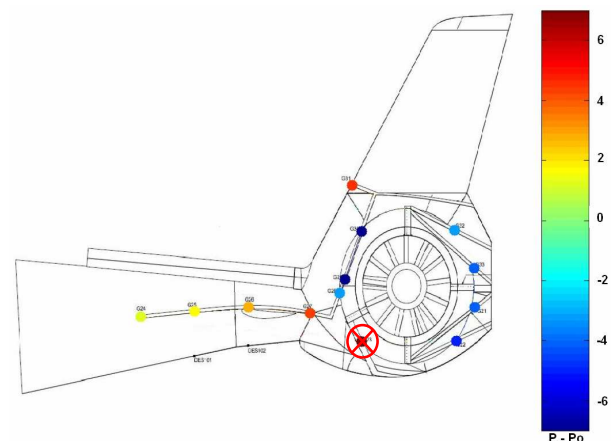
### 5.3.3. Static pressures around the tail boom

In order to measure the pressure around the rear part of the fuselage, flexible strips in silicon were

elaborated and stuck along the tail boom and around the Fenestron.

Each strip includes 8 ducts which were perforated at different locations, providing 13 pressure holes on both sides of the helicopter tail boom. Following figures (Figure 16-18) show each side of the helicopter tail boom. Silicon strips are represented as mounted on the helicopter and the 26 static pressures are symbolized by colored dots.

Figure 16 shows an average value of the static pressures measured along the tail boom during the entire first campaign. These values are the difference between the local pressure and the helicopter static pressure.

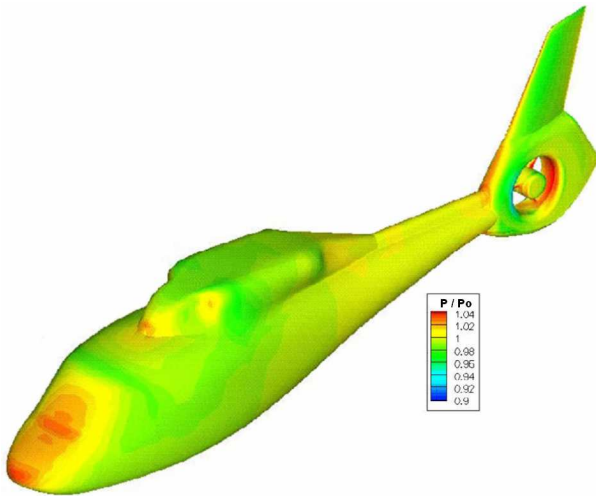


**Figure 16: Static pressure on the helicopter port side**

Thus, the pressure is increasing (yellow or red dots) along the tail boom. Another high pressure region is located on the leading edge of the fin. Around the Fenestron, a diminution of the pressure can be observed. One pressure hole provided very high pressure values during all flights revealing an incoherent behaviour. This pressure hole has been crossed out on the graph.

Figure 17, which corresponds to a CFD result issued from [4], illustrates a full fuselage computation of steady pressures for a level flight at 150 knots. A good correlation with preceding experimental data can be observed. The pressure distribution shows the location of the stagnation point on the nose of the fuselage for this configuration but also confirms the one on the leading edge of the fin. Another good correlation between experimental data and computations is shown along the tail boom, where the pressure is also increasing.

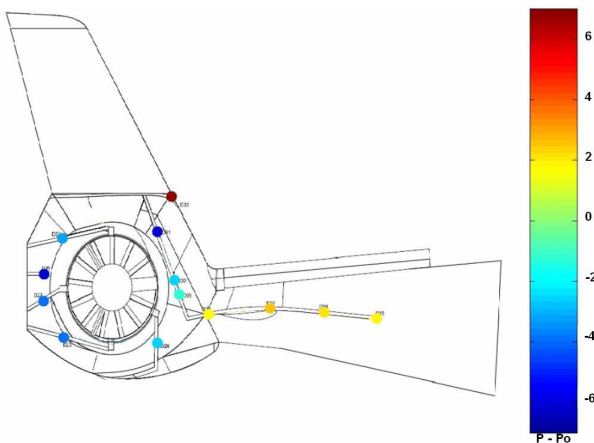




**Figure 17: Computed pressure distribution on the fuselage in level flight**

Finally, computations predicted in green areas a diminution of the pressure around the Fenestron vein which has been demonstrated in flight (Figure 16).

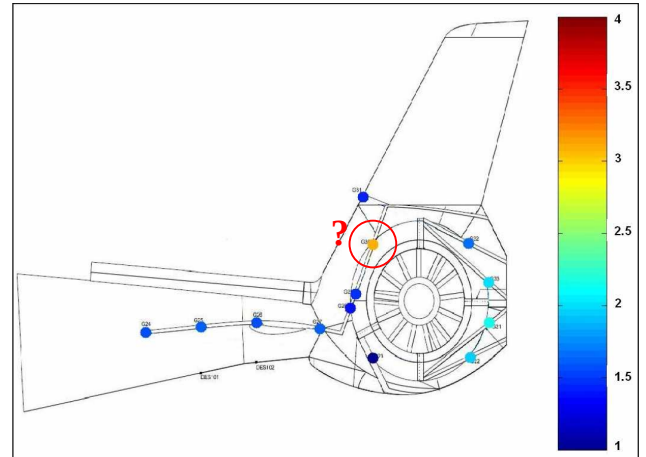
The next figure represents the static pressures measured along the tail boom on the helicopter starboard side. Same remarks as before can be made. The pressure is increasing along the tail boom and a stagnation point is visible on the leading edge of the fin (pressure hole different from the previous one on figure 16). Here again, the pressure around the Fenestron vein on the exit side is decreasing.



**Figure 18: Static pressure on the helicopter starboard side**

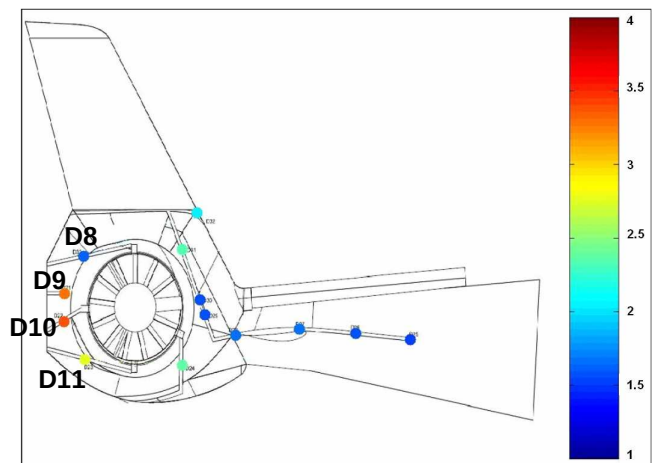
Following figures represent an average of the peak to peak variation of the pressure observed in all flight phases.

Figure 19 shows the helicopter port side. Values measured on the pressure holes represented in dark blue had a relative low standard deviation which would indicate a relative steady flow. On this side, one pressure tap reveals higher pressure variations but its location (red circle on the figure) cannot alone explain this behavior. That's why this channel has to be considered as suspicious.



**Figure 19: Peak to peak variation of the static pressures on the port side**

On the other side, hot colors indicating an increase of the peak to peak variation, three pressure taps show clearly higher pressure variations. These pressure taps are noted D9, D10 and D11 and are located on the exit side on the rear part of the fuselage.

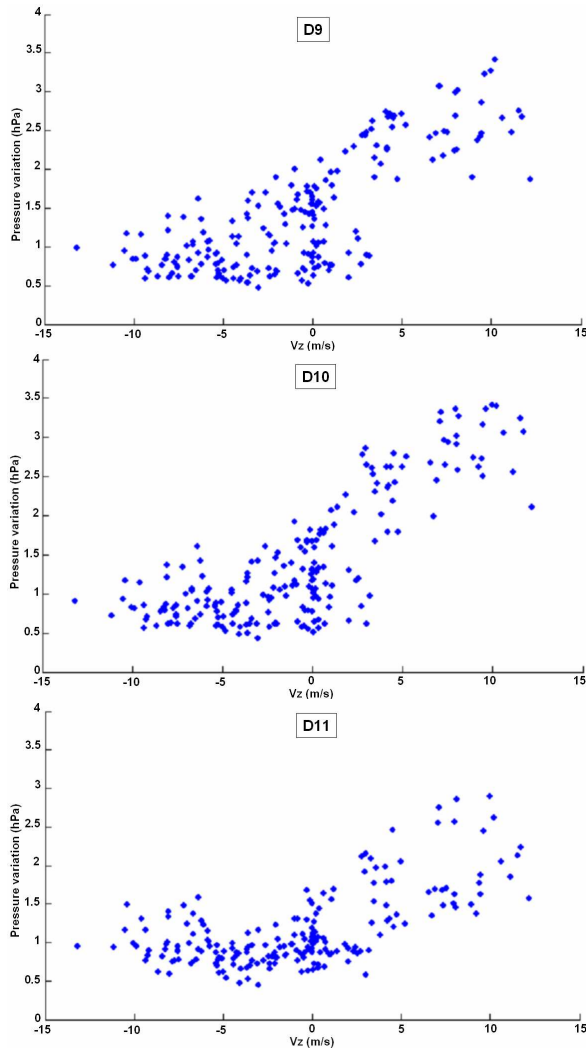


**Figure 20: Peak to peak variation of the static pressures on the starboard side**

This high level of pressure variations suggests a very perturbed flow in this part of the fuselage. The

pressure tap noted D8 placed just above presents low pressure variations which indicate a region of steady flow.

Moreover, on figure 21 is represented the average peak to peak variation of the three static pressures D9, D10 and D11 as function of the vertical speed Vz for all selected flight phases.



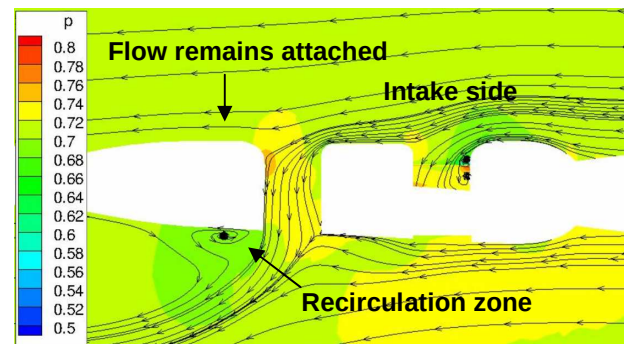
**Figure 21: Peak to peak variation of 3 static pressures on the starboard side function of the vertical speed**

The link between peak to peak pressure variations and vertical speed is obvious. It has to be mentioned that this relation is not observed on pressure tap D8 or any others.

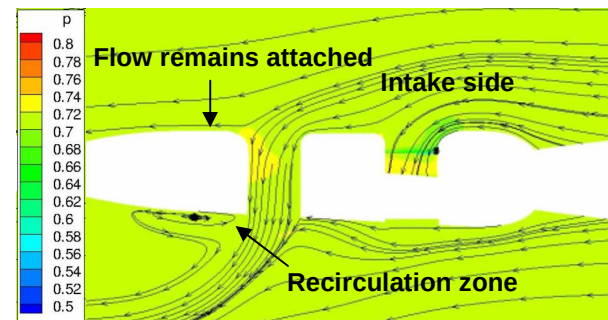
As shown before, the need of thrust lead to an increase of the tail rotor induced velocity with the rate of climb. As a result, the velocity of the flow exiting the Fenestron duct is higher in climb. Mixed

to the outside airflow, this generates a recirculation zone on the rear part of the fuselage which is clearly visible on CFD computations.

The following figures (Fig. 22 and 23) show the surface restricted streamlines and pressure contours obtained by CFD computations in a horizontal cut of the Fenestron at mid-vein for two different flight cases. First one is a level flight at 150kts, second one at 75kts with a rate of climb of +1500 ft/min. In both cases, on the intake side of the Fenestron, the flow remains attached to the rear part of the tail whereas a recirculation zone is established on the exit side. In climb, this recirculation zone seems wider. Considering their locations, pressure taps D9, D10 and D11 are plunged in a recirculation zone which explains a high level of pressure variation, in all flight phases, but with an important increase in climb.



**Figure 22: CFD computations: Level flight at 150 kts**



**Figure 23: CFD computations: Climb at 75 kt, Vz = +1500 ft/min**

#### 5.3.4. Static pressures in the Fenestron vein

The following figure (Figure 24) is a closed view of the downstream part of the Fenestron vein in level flight at 150 kts. The pressure distribution shows the diminution of the pressure in the part of the Fenestron vein located upstream (blue zone) and the increase in the part of the vein located downstream (red zone). We can also see the

stagnation area on the hub support, as well as on the leading edge of each of the three stator blades and on the leading edge of the fin. An image of the real helicopter is shown in the top right corner of the figure where it can be seen the generating line including 20 pressure taps. Figure 25 represents this generating line and colored dots are an average value of the static pressures (referenced to the helicopter static pressure  $P_o$ ) measured in all flight phases. On the right part of this generating line, from pressure taps noted 1 to 9, the pressure is decreasing along the fuselage which indicates an acceleration of the flow ingested by the Fenestron. But in the vein, the pressure re-increases from the beginning of the upstream part to the rotor plan and behind. Finally, the flow expands to the atmospheric pressure on the exit side of the Fenestron.

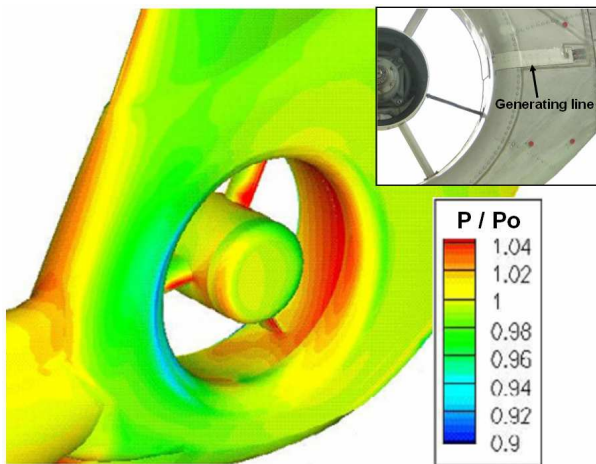


Figure 24: CFD computations: Downstream view

Following figure (Figure 26) is the CFD closed view of the upstream part of the Fenestron vein in the same configuration. On the right bottom, an image of the generating line mounted on the helicopter.

As before, figure 27 represents the generating line and the average value of the static pressures. On the right part of this figure, the pressure taps placed on the upstream part of the Fenestron vein show a gradual decrease of the pressure as they are closed to rotor plan, which corresponds to the CFD green areas. The 10<sup>th</sup> pressure tap was considered out of order.

Behind the rotor plan, in the Fenestron diffuser, the pressure re-increases as forecasted by CFD computations.

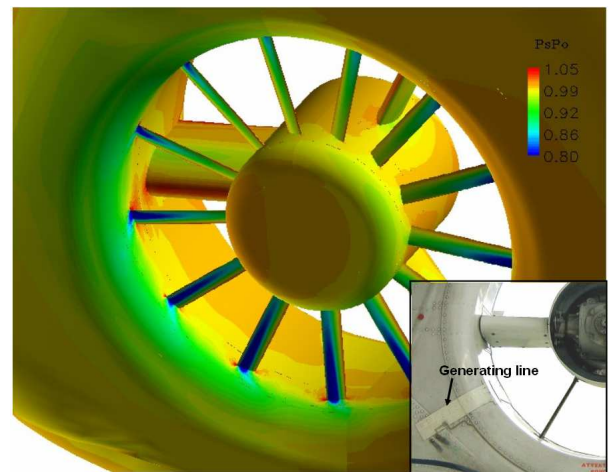


Figure 25: CFD computations: Upstream view

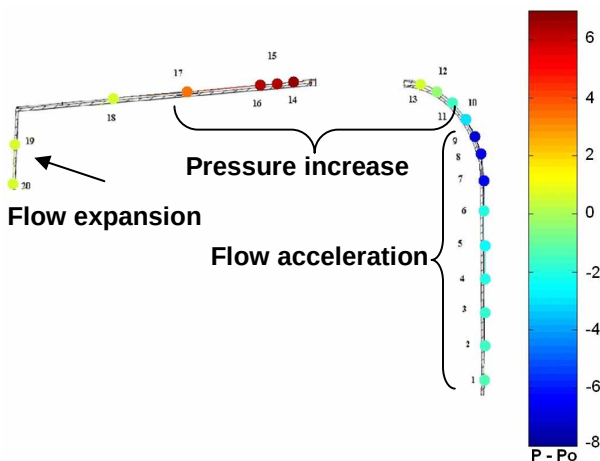


Figure 26: Downstream generating line with 20 static pressure taps

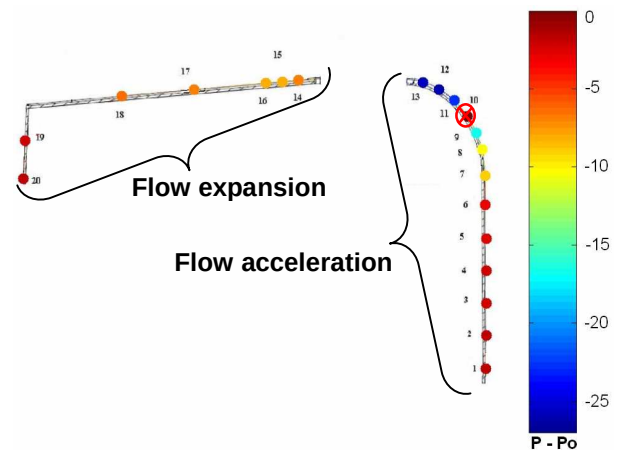


Figure 27: Upstream generating lines with 20 static pressure taps

### 5.3.5. Unsteady pressure measurements

Figure 28 shows the difference of the unsteady pressure measured between upper and lower surfaces of two blades. All these measures are the average values observed in all stabilized phases. Large differences between the pressures measured on the two surfaces of the blades are mainly observed in climb between 55 and 90 kts. In those conditions, the tail rotor is highly loaded. The difference reaches 87 hPa instead of 20 hPa in descending or level flight. These data are not surprising. The tail rotor thrust is obtained by this difference of pressure and the tail rotor thrust needed is higher in climb. These differences

decrease with the increase of the forward speed and the rate of descent, the fin becoming more and more efficient and the tail rotor thrust being considerably reduced.

As mentioned before, the Fenestron works in a reverse mode in steep descent. Moreover, during this campaign, these steep descents were performed with no side slip angle, leading the pilot to further unload the tail rotor. This decrease of required thrust in descending flights can be observed when the difference of pressure between the two surfaces is drastically reduced, in red regions on the figure.

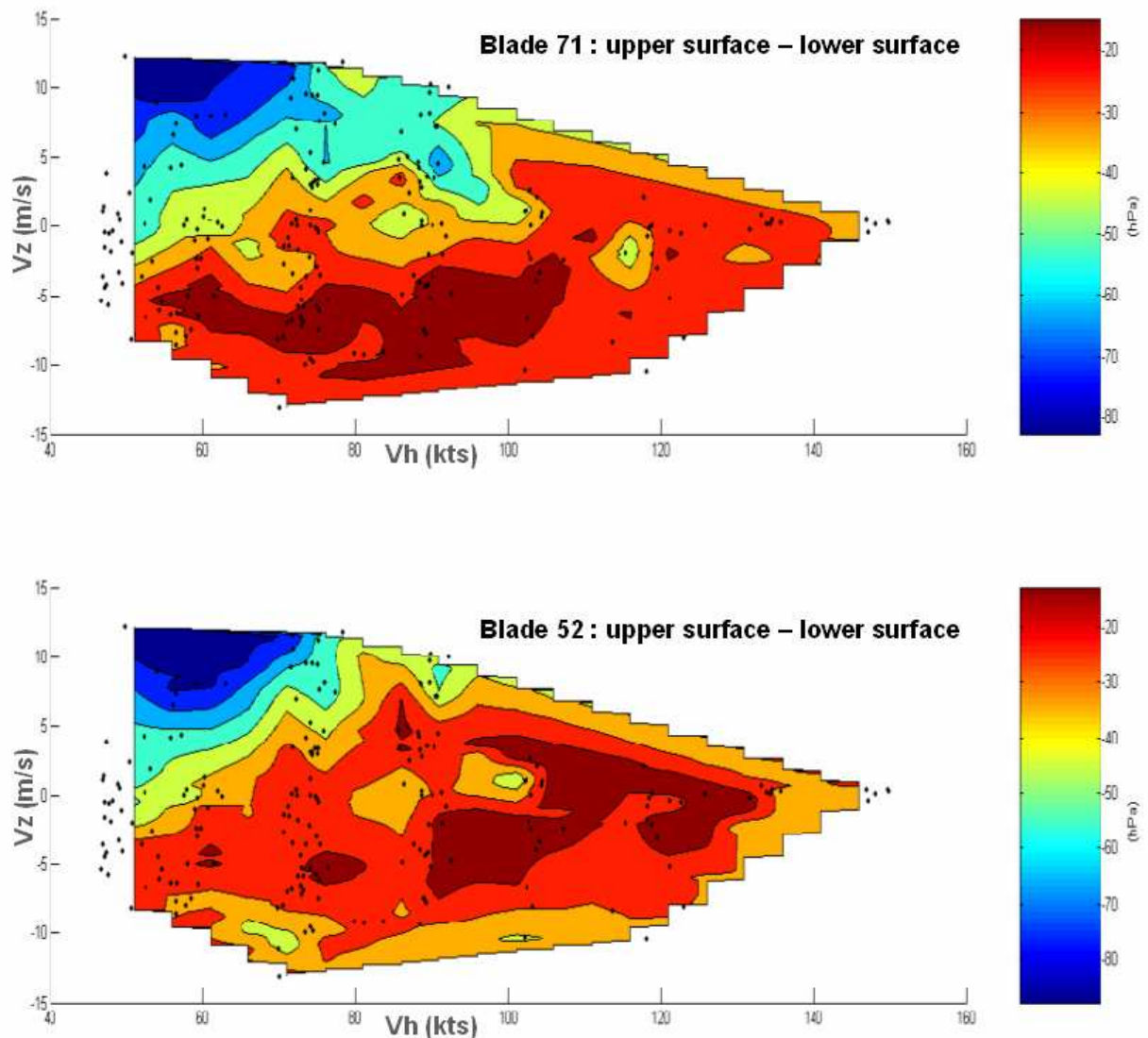
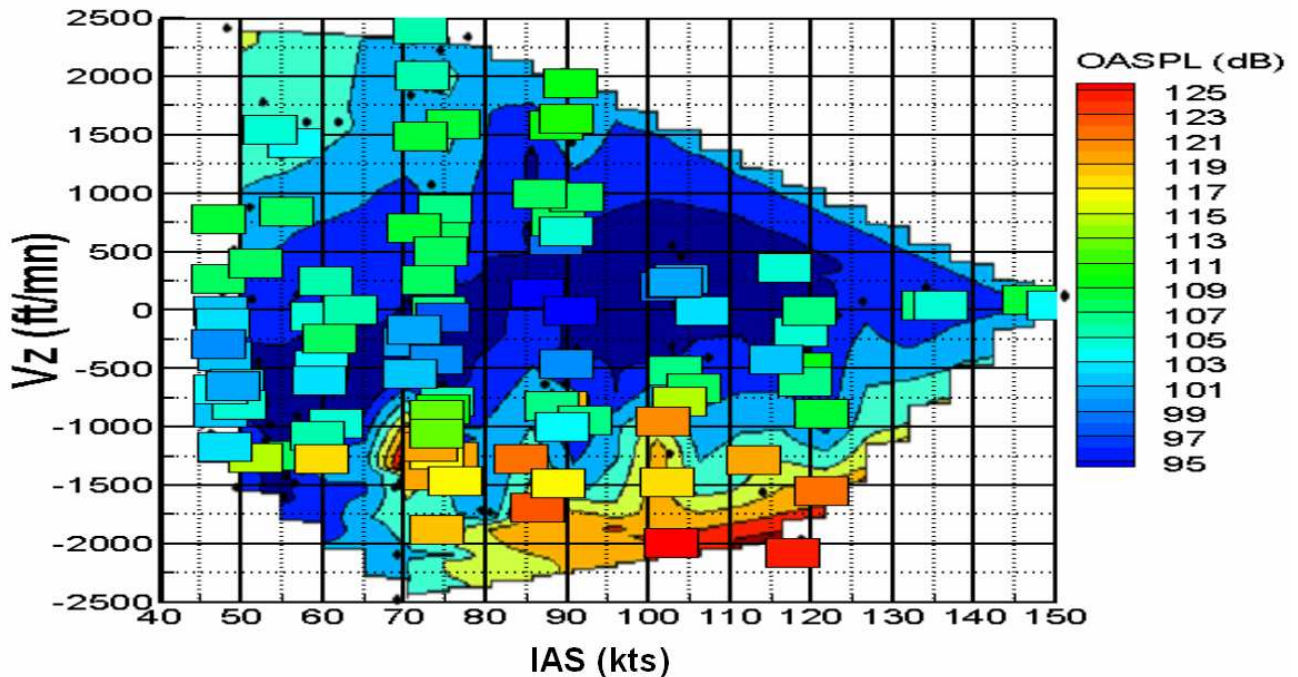


Figure 28: Pressure difference between upper and lower surfaces of two blades





**Figure 29: Correlation between unsteady pressure peak to peak variations measured in the vein and recorded noise**

On Figure 29 are superposed, as a function of the flight conditions, the unsteady pressure peak to peak variations observed on one Kulite in the vein and the OASPL of one of the microphone located on the left empennage.

The highest OASPL values are reached for steep approaches at high airspeeds. As discussed before, the Fenestron thrust needed to counterbalance the main rotor torque is particularly low in these conditions. This result has already been highlighted in papers [2] and [4] where it was shown that there's no direct relationship between tail rotor thrust and radiated noise. Particularly, flight conditions with lower Fenestron blade loading lead to increased radiated noise, because of specific acoustic blade-to-blade interferences.

Moreover, as the Fenestron is working in a reverse mode in these conditions, the flow inside the duct is highly perturbed which would leading to high pressure variations in the vein and high load variations on the blades. As a matter of fact, this second assumption was verified through the analysis of the spectra of the blade load in paper [2]. In the vein, the demonstration of a disturbed flow is given by the significant increase of the pressure variation in steep descent between 90 and 120 kts and for sink rates lower than -800 ft/min.

Low levels of pressure peak to peak variation are noticed in level flight, at any flight speed. A very slight increase can be observed in climb at moderate speeds below 60 kts which corresponds to flight cases at high level of tail rotor thrust and blade loading. In these cases, the flow inside the vein is certainly more homogeneous and cannot alone explain these variations of pressure. On the other hand, Kulites were placed closed to the blades plan and these variations could be a consequence of the blades passing.

### 5.3.6. Blades strains

First dedicated to the real time survey of the strains on the blades, the strains gauges have been analyzed and interesting results can be presented.

The next figure (Figure 30) represents the static strains plotted in a  $V_h/V_z$  chart. All these measures are the average values observed in all stabilized phases.

High static strains are mainly observed in climb between 55 and 90 kts. As the tail rotor is highly loaded in those conditions, high static strains on the blades are not surprising.

Here again, in steep descent or at high forward speed, as the thrust needed is decreasing, the strains on the blades are also considerably reduced.

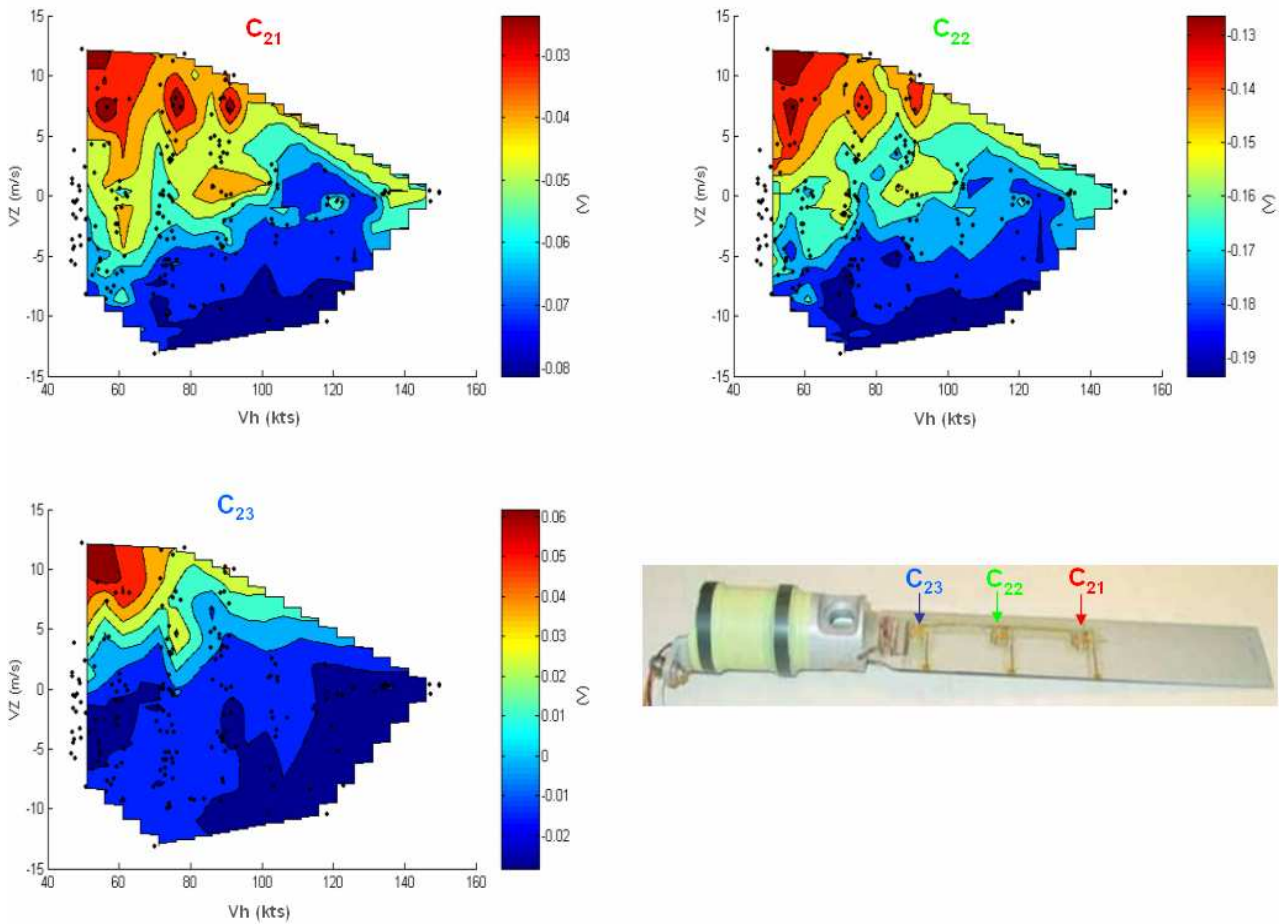
Experimental data confirmed that the flow inside the Fenestron can be considerably perturbed in some flight conditions.

One consequence of a perturbed flow in the Fenestron duct is the alternated strains observed on the blade. Indeed, analyzing the recorded data from the blade gauges revealed periodic peaks on the strains as shown in figure 31. In this example, the time between two peaks is around 12.91 ms, corresponding to a rotor RPM of about 4645.

Figure 32 shows CFD results in a horizontal cut of the Fenestron at mid-vein where several recirculation areas can be noticed but the first and larger one is located at the upwind entrance of the vein (just in front of the tail rotor drive support). This recirculation

area can generate aerodynamic disturbances on the blades but it is closely dependent of the flight case.

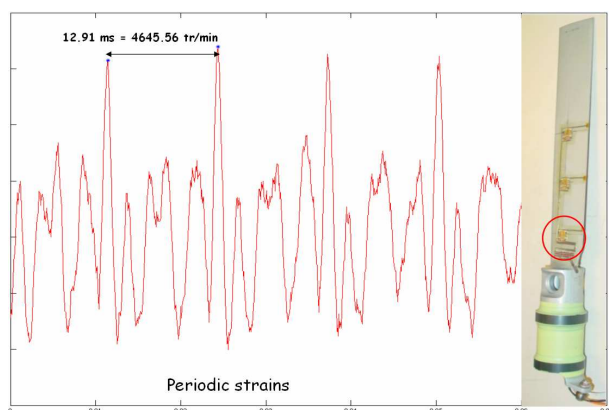
However, this phenomenon was observed in all flight cases independently of the loads applied on the rotor and also in climb, flight case for which this recirculation area is extremely reduced as foreseen by CFD computations and previously shown on figure 23. This led us to the conclusion that this rather high level of dynamic loads is certainly due to the passage of the blade in front of the tail rotor drive support. Indeed, the tail rotor drive support is a rather large piece of metal placed in the Fenestron duct which probably creates locally a very disturbed airflow causing blade flapping and then, alternated strains.



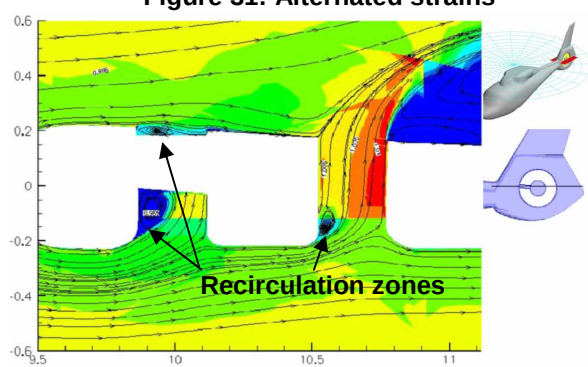
**Figure 30: Static strains measured on the blade**

Nevertheless, the accurate position of the blade when the peak occurs is not known. This “physical” RPM counter couldn’t be used in the first campaign to synchronize the fixed and rotating measurement chains with the required precision but allowed us to

verify the lack of transmission delays on the telemetry stream.



**Figure 31: Alternated strains**



**Figure 32: Tail rotor drive in Fenestron duct**

Repair of the synchronizing pulses was undertaken for the second flight campaign and should enable to localize the blades in the rotating frame. This should help us to know the accurate position of the blade when the peak occurs and to better understand the reason of these alternated strains.

## 6. CONCLUSIONS

A very innovative instrumentation and measurement chain architecture has been designed, achieved and tested by the Department of System Controls and Flight Dynamics of ONERA, Eurocopter and the CEV. Such a flight test campaign with a fully instrumented Fenestron including a unique rotating telemetry system is a world premiere.

The entire first flight test program was successfully performed in eight flights from October 2008 to January 2009. As a result of a five years' work, this first flight test campaign provided a fairly complete aerodynamic and acoustic database relative to the flow and noise mechanisms in the Fenestron of the Dauphin 365N.

This rather large database should enable to improve the current Fenestron models in flight mechanics codes. For example, it has been noticed that the two Fenestron models present in the HOST seems to under-estimate the induced velocity. The impact of such an under-estimation of the total power required for hover and climb have highlighted the need of such a tuning.

Moreover, the impact of exhaust gases on the Fenestron has been noticed. Thanks to the HOST code, it reveals that it represents a low modification of the power required depending on flight conditions. But, it has to be taken into account such as other loss of power factors and the development of such a modeling is in progress.

From a qualitative point of view, comparisons have been made between static pressure measured on the fuselage and in the Fenestron vein and CFD computations showing good agreements. This work has also been undertaken in the ONERA Department of Applied Aerodynamics for CFD code validation.

Some parameters reveal that the flow in the Fenestron vein can be highly perturbed in some flight conditions. Aerodynamic measurements as well as unsteady pressures confirmed a complex flow field in the vein, especially in steep descent flights. These conditions correspond to a reverse functioning mode of the tail rotor for which the highest noise level are reached.

Finally, first dedicated to the real time survey of the strains on the blades, strains gauges have also been analyzed. Static strains only revealed a good agreement with the flight conditions as a function of the loads required on the blades. Moreover, focusing on dynamic strains, a peak was observed every rotation. After exploring different possibilities, we came to the conclusion that this rather high level of dynamic loads is certainly due to the passage of the blade in front of the tail rotor drive support. For that goal, we look forward to checking the good functioning of the two synchronization pulses.

As mentioned before, the first campaign provided a unique experimental database enabling aeroacoustic code validation. These codes could then be used to improve the shape of the next generation of Fenestron. Flight mechanics codes Fenestron models will also take benefits of such a

database. In addition, the second flight test campaign will provide a broad database of acoustic footprints correlated to on-board measurements. Comparisons with new aeroacoustic simulations will be provided by aerodynamics and aero-acoustics departments of ONERA in further papers.

This resulting database should help to develop quieter helicopters by providing validated tools to define low noise flight procedures or to improve the shape of the new generations of Fenestron. In that sense, this research study perfectly corresponds to the general context of the European Clean sky's Green Rotorcraft research platform or Friendcopter program, whose aims are to develop helicopters more passenger and environmentally friendly.

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